

Exercise series 5: SQL and Aggregation



ADReM
Advanced Database Research & Modelling
University of Antwerp

 **Universiteit
Antwerpen**

How Aggregation Works

Find the makers who don't make printers and make more than one model of computer together with the number they make.

```
SELECT maker, count(*)  
FROM Product P  
WHERE maker NOT IN  
(  
    SELECT maker  
    FROM Product  
    WHERE type = 'printer'  
)  
GROUP BY Maker  
HAVING count(model) >= 2  
;
```

Product		
maker	model	type
'A'	1001	'pc'
'A'	1002	'pc'
'A'	1003	'pc'
'A'	2004	'laptop'
'A'	2005	'laptop'
'A'	2006	'laptop'
'B'	1004	'pc'
'B'	1005	'pc'
'B'	1006	'pc'
'B'	2007	'laptop'
'C'	1007	'pc'
'D'	1008	'pc'
'D'	1009	'pc'
'D'	1010	'pc'
'D'	3004	'printer'
'D'	3005	'printer'
'E'	1011	'pc'
'E'	1012	'pc'
'E'	1013	'pc'
'E'	2001	'laptop'
'E'	2002	'laptop'
'E'	2003	'laptop'
'E'	3001	'printer'
'E'	3002	'printer'
'E'	3003	'printer'
'F'	2008	'laptop'
'F'	2009	'laptop'
'G'	2010	'laptop'
'H'	3006	'printer'
'H'	3007	'printer'

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'A'	1001	'pc'
'A'	1002	'pc'
'A'	1003	'pc'
'A'	2004	'laptop'
'A'	2005	'laptop'
'A'	2006	'laptop'
'B'	1004	'pc'
'B'	1005	'pc'
'B'	1006	'pc'
'B'	2007	'laptop'
'C'	1007	'pc'
'D'	1008	'pc'
'D'	1009	'pc'
'D'	1010	'pc'
'D'	3004	'printer'
'D'	3005	'printer'
'E'	1011	'pc'
'E'	1012	'pc'
'E'	1013	'pc'
'E'	2001	'laptop'
'E'	2002	'laptop'
'E'	2003	'laptop'
'E'	3001	'printer'
'E'	3002	'printer'
'E'	3003	'printer'
'F'	2008	'laptop'
'F'	2009	'laptop'
'G'	2010	'laptop'
'H'	3006	'printer'
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'A'	1002	'pc'
'A'	1003	'pc'
'A'	2004	'laptop'
'A'	2005	'laptop'
'A'	2006	'laptop'
'B'	1004	'pc'
'B'	1005	'pc'
'B'	1006	'pc'
'B'	2007	'laptop'
'C'	1007	'pc'
'F'	2008	'laptop'
'F'	2009	'laptop'
'G'	2010	'laptop'

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	1002	'pc'
	1003	'pc'
	2004	'laptop'
	2005	'laptop'
	2006	'laptop'
'B'	1004	'pc'
	1005	'pc'
	1006	'pc'
	2007	'laptop'
'C'	1007	'pc'
'F'	2008	'laptop'
	2009	'laptop'
'G'	2010	'laptop'

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maker	model	type
'A'	1001	'pc'
	1002	'pc'
	1003	'pc'
	2004	'laptop'
	2005	'laptop'
	2006	'laptop'
'B'	1004	'pc'
	1005	'pc'
	1006	'pc'
	2007	'laptop'
'F'	2008	'laptop'
	2009	'laptop'

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```

Product	
maker	count(*)
'A'	6
'B'	4
'F'	2

How Aggregation Works

Find the makers that produce exactly two different types of product

```
SELECT maker  
FROM Product P  
GROUP BY Maker  
HAVING count(distinct type) = 2  
;
```

Product		
maker	model	type
'A'	1001	'pc'
'A'	1002	'pc'
'A'	1003	'pc'
'A'	2004	'laptop'
'A'	2005	'laptop'
'A'	2006	'laptop'
'B'	1004	'pc'
'B'	1005	'pc'
'B'	1006	'pc'
'B'	2007	'laptop'
'C'	1007	'pc'
'D'	1008	'pc'
'D'	1009	'pc'
'D'	1010	'pc'
'D'	3004	'printer'
'D'	3005	'printer'
'E'	1011	'pc'
'E'	1012	'pc'
'E'	1013	'pc'
'E'	2001	'laptop'
'E'	2002	'laptop'
'E'	2003	'laptop'
'E'	3001	'printer'
'E'	3002	'printer'
'E'	3003	'printer'
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	1003	'pc'
	2004	'laptop'
	2005	'laptop'
	2006	'laptop'
'B'	1004	'pc'
	1005	'pc'
	1006	'pc'
	2007	'laptop'
'C'	1007	'pc'
'D'	1008	'pc'
	1009	'pc'
	1010	'pc'
	3004	'printer'
	3005	'printer'
'E'	1011	'pc'
	1012	'pc'
	1013	'pc'
	2001	'laptop'
	2002	'laptop'
	2003	'laptop'
	3001	'printer'
	3002	'printer'
	3003	'printer'
'F'	2008	'laptop'
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'G'	2010	'laptop'
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	1003	
	2004	'laptop'
	2005	
	2006	
'B'	1004	'pc'
	1005	
	1006	'laptop'
'C'	2007	
	1007	'pc'
'D'	1008	'pc'
	1009	
	1010	
	3004	'printer'
	3005	
'E'	1011	'pc'
	1012	
	1013	
	2001	'laptop'
	2002	
	2003	
	3001	'printer'
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	1010	
	3004	'printer'
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GROUP BY Maker  
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```

Product maker
'A'
'B'
'D'